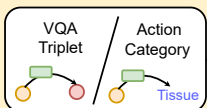
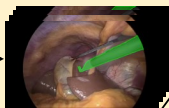


CholecSeg8k / GraSP

Clip
Selection

selection



1-10s

EndoVis-17 / EndoVis-18

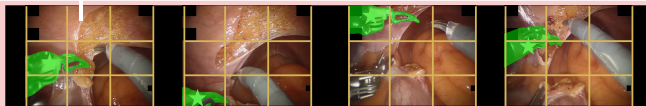
Pixel Motion



selection



1-10s

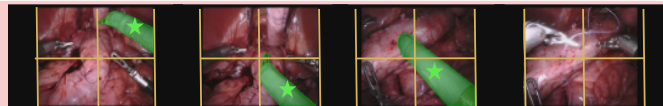
Image
Partitioning

on the mid-left

on the bottom-left

on the top-left

on the mid-left



on the right

in the bottom

on the right

disappear

Expression
Generation

Reformulate the {Bipolar forceps, travel, tissue} with {spatial information} into a natural language description of motion and interaction over time

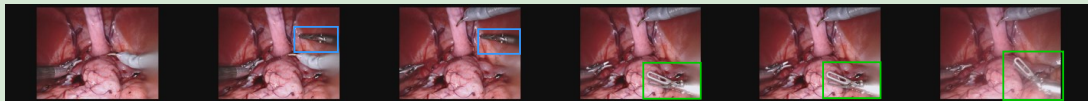


Bipolar forcep travels from **mid-left** to **bottom-left** while closing, then hold tissue at the **top left**, finally pull the tissue to the **mid-left**

Monopolar curved scissors + spatial info

Manual annotation

Monopolar curved scissors is cauterizing tissue, **moving up and down on the right**, and **disappearing**.

Final
DataManually
Check

Monopolar curved scissor **show up and disappear at the top-right**

Prograsp forcep is **idle on the bottom-right** and then **lifted in the same place**